Energy Price Prediction using Transformer Architecture

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Abstract. Nothing here yet

1 Introduction

Energy Price Forecasting is a highly sought after technique that adds value to companies that benefit from load balancing and management. With the further decarbonisation of the energy grid, renewable energy introduces a strong temporal characteristic to the challenge; one that Neural Networks, particularly advancements in RNN[1, 20, 15, 4, 10], Transformer[8, 18, 12, 19] and ensemble[2, 3, 6] technology can leverage to improve the accuracy of predictions. As feature extraction mechanisms grow more robust[3, 1, 20, 13, 15], and models involve complex sets of features like weather[14, 18], location, and energy consumption[13], the demand for better, more accurace, and more farreaching models is growing withing the industry.

One of the industry players with a keen interest in the technology is OpenRemote. The company provides a platform where entities with multiple energy-demanding components can manage their resources. One of the microservices that OpenRemote provides is an Energy Price Forecasting (EPF) model hosted by Prophet, that does not employ Neural Networks for its predictions[17]. Their current service is only able to offer 24h predictions ahead of time, which causes constraint for schedueled load balancing, and limits the amount of saving a client could make. OpenRemote argues that a 48h prediction window would be more benefitial. Furthemore, the Prophet model is more accurate around the peaks, but relatively innacurate anywhere else. This is due to the limitations of the modeling techniques employed which, although easy to configure and faster than a Neural Network to train and generate forecastings, struggles with the complexities of multi-feature temporal and spacial data that RNNs and Transformers capture with greater ease.

This paper will present, out of a collection of methods collected via a thourough literature review, the results of experiments conducted with the intention of comparing different combinations of methods - both preprocesing, and neural network architectures. The end result if the analysis and thourough review of the results. It will also provide advice on how to implement the techniqeus reviewed throughut this research.

2 Methods

The objective of this research is to interrogate a variety of methods within the literature surrounding time-series regression. More specifically, energy price or energy related time-series prediction. This particular research field adopts a multi-feature, and often multimodal approach to the problem. This section outlines the methods used for the research, in hopes of finding a model architecture that outperforms the current method that OpenRemote employes.

2.1 Research Design

The research is designed to almost perform as a census on the available methodologies. A healthy variety of methods for preprocessing and architectures were choosen and divided into experiments that compile similar architectures, with sub experiemnts which introduce small variations. These variations intend to reveal the effect of a particular method i.e. if the difference between experiment 1.a and 1.b is one specific architecture, then the difference in performance might have to do with the introduction of that architecture.

The scope of the research is strictly for time-series energy price prediction, and its goal of being more accurate than the benchmark.

We will, for the purposes of this proposal, use a naming convention of Exp.$i.v$, where i is the number of the experiment configuration, and v is a letter, whose order is reset for each i.

	Exp.1			
	a	b	c	d
Architecture	CNN-BiLSTM	CNN-xLSTM	CNN-BiLSTM-Transformer	CNN-Transformer
Preprocessing methods	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST
Additional notes				To prove null-hypothesis i.e. RNNs punish accuracy

Table 1: Exp.1: Model Architectures and Preprocessing

2.1.1 Exp.1

From our literature, one of the most recurrent topics is the power of ensemble models, particularly ones using RNNs. To start out simple, we will introduce a CNN-LSTM variant architecture as a base and design multiple experiments that build upon it.

Since we are using CNNs, which serve the purpose of capturing long-term sequential patters, it would be ideal to test an array of preprocessing methods, some of which enhance 3-dimentional features. For Exp.1, all variations will cycle through:

1. Normalisation
2. Wavelet Transform[13, 5]
3. Transformer Encoding (PatchTST[11])

Exp.1.a Variation **a** will use a simple CNN-BiLSTM, since the use of BiLSTM consistently outperforms normal LSTMs in time-series forecasts.

Exp.1.b Variation **b** will use a CNN-xLSTM.

Exp.1.c Variation **c** will use a CNN-BiLSTM-Transformer. The discrepancy between results for each preprocessing method should be greatly scrutinised for this variation, as the use of a Transformer encoder and output model essentially places the CNN and BiLSTM as feature extractors.

Exp.1.d Variation **d** will use a CNN-Transformer. By removing the transformer, we aim to prove the null hypothesis that RNNs, particularly LSTMs, are greatly outperformed by Transformers for short-term dependencies, and that the features that they extract act as noise for a transformer.

This set of variations offer a comprehensive use of the methods that the literature has provided, but they also take into account standard practises i.e. Normalisation and the use of CNNs for time-series data, also found in the literature[2, 20, 10].

2.1.2 Exp.2

Current trends in AI strongly favour Transformer architectures, or Attention layers. Exp.2 will focus on exploring this particular aspect in more detail. Exp.1 also features Transformers, but in this section, we want to amplify their strenghts by means of expanding our number of features. Non-linearity is accutely endemic to all of our features[10]. One hypothesis is that the Transformer architecture will be able to capture linear features where there is usually non-linearity.

Exp.2.a Variation **a** will use a classic multi-head Decoder-Only transformer[12]. Decoder-Only transformers perform well in renewable scenario analysis, and use less memory[6]. This is relevant for our study because the nature of the data and analysis is similar by virtue of the non-linear and complex time-series tasks both try to address.

Exp.2.b Variation **b** will use the same model as Exp.2.a, but utilise Wavelet Transformers for feature extraction[21].

Exp.2.c Variation **c** will use a multi-head Encoder-Decoder Transformer. Encoder-Decoder variants perform great with forecasting tasks[6].

Exp.2.d Variation **d** will use a Kernel-based Transformer. This architecture can better capture linearity when it exists[16] which has been pointed out as a reason for transformers failing to generalise on Time-Series tasks[7] , all the while using less memory.

Exp.2.e Variation **e** will use an inverted Transformer, henceforth known as iTransformer; a method introduced by a landmark paper back

	Exp.2				
	a	b	c	d	e
Architecture	Decoder-Only Transformer	Decoder-Only + FFT	Encoder-Decoder Transformer	Kernel-based Transformer	iTransformer
Preprocessing methods	Standard	Wavelet Transform	Standard	Kernel-based	Standard
Additional notes		Same model as **a**			

Table 2: Exp.2: Model Architectures and Preprocessing

in 2023[9]. The paper specifically demonstrates the effectiveness of the proposed methodology for Time-Series Forecasting, and has been explicitely used for EPF[18].

2.2 Data

The dataset comprehends a meryad of features from different sources, such as metereological data, overall energy consumption and production, and gas prices. The target variable, being the energy price, is also included. The $\vec{\Delta}$ between samples is 1 hour.

The data is also preprocessed using the methods above. Timestamp data is feature engineered to decompose it into many features, such as hour, day of week, month, quarter and weekend, all with cyclical scaling.

2.3 Procedure

All experiment models are based on the Torch library. For the preprocessing, there is a mixture of self-written methods, and some from scikit-learn and pywt, for Wavelet transform. They are arranged into the experiment configurations described above, and run sequentially using a custom script, that outputs the results in bulk.

2.4 Preprocessors

The preprocessors used in this experiment merit a section of their own, as they are an integral part of the research, and overlwhelm the project with a veneer of complexity alike the ensemble models described above. We will go through the main preprocessing methods, and explain what they constitute; provide a logical explanation for the methodology employed. Some preprocessors might discrimitate the experiment that they belong to.

2.4.1 Standard Preprocessor (EXP. 2)

The standard processor applies simple processing on the data. For floating points, it applies a standard scaler, and a MinMax scaler. For datetime data, it applies Cyclical Scaling.

Standard Scaler The standard scaler simply uses Standard Normal Distribution to normalise the values. This helps with uneven scales within the dataset. So the mean of each column becomes 0 and the standard deviation becomes 1. The MinMaxScaler scales the data in a max to min scale. So the minimum value is 0 (or -1 if there are negative values for that feature) and the max value becomes 1.

Both are used because, while standard scaling is useful for baseline detection, the MinMax Scaler ensures that we dont lose outlier feedback.

Cyclical Scaling This scaling technique is used for date and time, and its purpose is to enforce the circular dimention of time onto the data, using trigonometry, as presented in 1. This is one of the valid methods for processing date and time columns for time-series data. This particular method provides cyclicality to time. Like how months, weekdays and hours are not linear but cyclical; they 'reset' on every cycle. This method hightlights that property. Thusly, we extract more features out of the original date.

Discrete Wavelet Transform A Discrete Wavelet Transform preprocessor is used for decomposing the signal into more nieche frequencies. This technique allows for nitpicking which "bands" or "signals" we want to remove, either because they capture a patter we want to obscure, or are just noise that the model can do nothing with.

Initially, we wrote down in the proposal that we would use Fast Fourier Transform, but for time-series data, the wavelet transform is more useful. The amplitude of the signal is not as useful as the decomposition of said signal, since it will allow us to reduce noise for some features, while preserving the overall patterns.

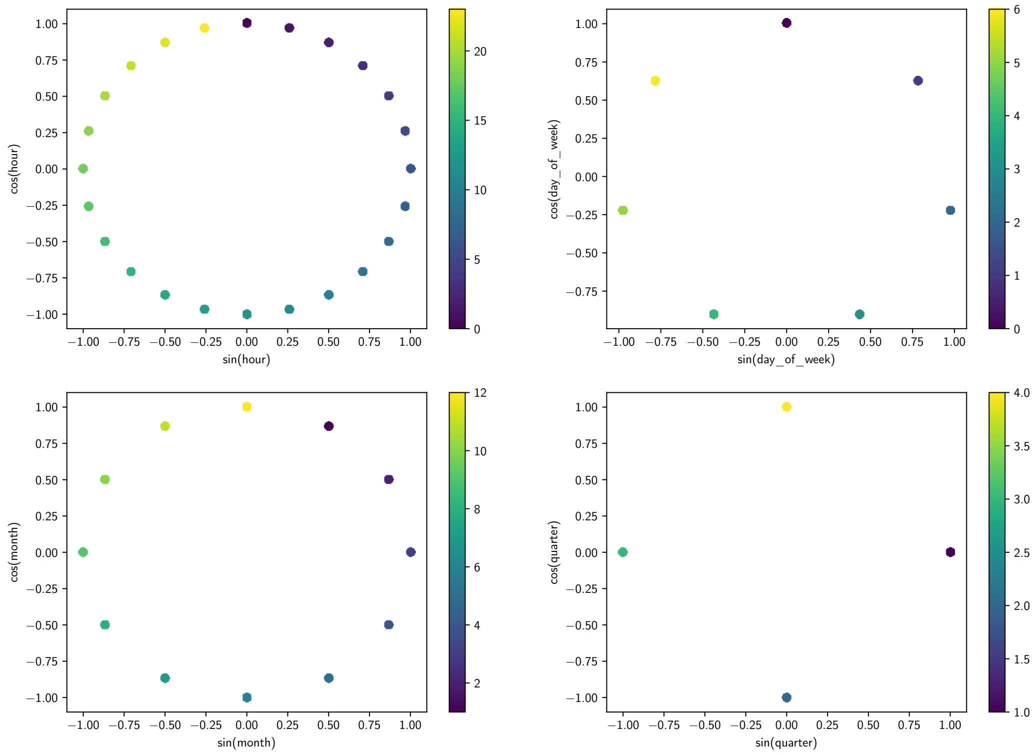

Figure 1: A plot of date features after being subjected to Cyclical Scalling

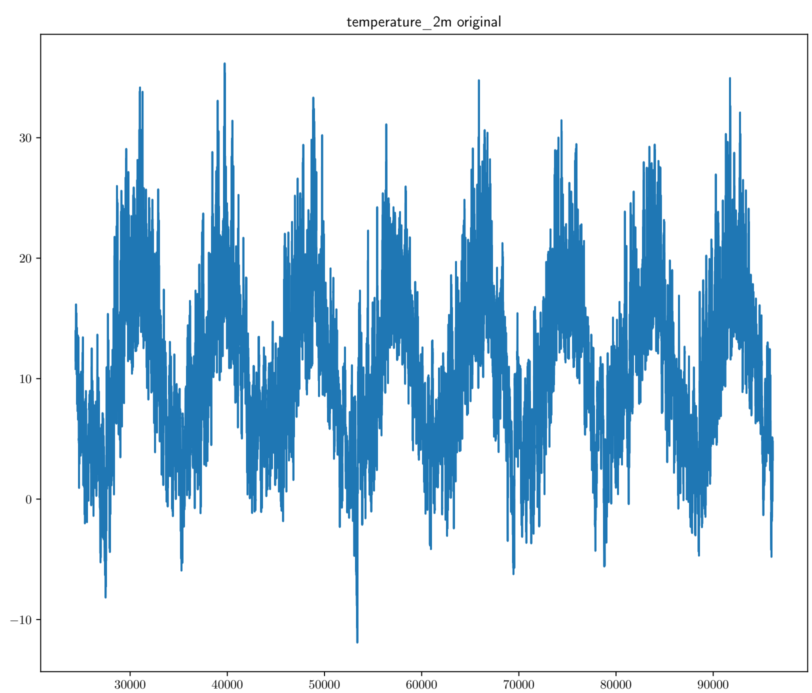

Figure 2: A plot of the temperature over time in our dataset

Consider Figure 2, where we can see in its totality, the temperature data in our dataset. Notice how there are minor imperfections in an otherwise clear periodical fluctuation. That is to be expected, as the Netherlands follows a pretty typical temperature cicle.

Now take a look at Figure 3. The signal from earlier is now decomposed into 3 different signals, called bands. Band 1 (or d1) looks like a lot of noise, same as Band 2. However, Band 3 seems a little more constrained. Altough it clearly detects peaks, it also finds stability. That is what we are looking for when doing Wavelet analysis.

Furthermore, by performing ablation[1], we are able to get a clearer picture of the effects of each band to a potential model.

variable	band	rmse	delta
total_load	d2	33.756189	0.249319
generation_forecast	d1	33.642795	0.135925
price	d1	33.632644	0.125774
temperature_2m	d3	33.614782	0.107912
Low	d2	33.605844	0.098975
wind_speed_10m	d3	33.603512	0.096642
Price	d3	33.599684	0.092814
price	d2	33.589829	0.082959
cloud_cover	d2	33.589608	0.082739
total_load	d3	33.579247	0.072377
temperature_2m	d2	33.574455	0.067585
shortwave_radiation	d2	33.569995	0.063125
High	d2	33.561575	0.054705
total_load	d1	33.552758	0.045888
Change %	d2	33.546079	0.039209
shortwave_radiation	d1	33.543296	0.036426
generation_forecast	d3	33.535409	0.028539
Low	d3	33.533773	0.026903
Change %	d3	33.532598	0.025728
cloud_cover	d3	33.520752	0.013882
wind_speed_10m	d2	33.519621	0.012751
High	d3	33.517491	0.010621
High	d1	33.515562	0.008692
Low	d1	33.513244	0.006374
Price	d1	33.508143	0.001273
cloud_cover	d1	33.501218	-0.005652
temperature_2m	d1	33.500676	-0.006193
shortwave_radiation	d3	33.496145	-0.010725
generation_forecast	d2	33.475815	-0.031055
Price	d2	33.473631	-0.033239
price	d3	33.470569	-0.036301
wind_speed_10m	d1	33.422056	-0.084814
Change %	d1	33.381141	-0.125729

Table 3: A table that shows the results of the ablation test per band and feature individually

We can read table 2.4.1 by simply looking at the column delta and understanding that the bigger the number is, the more benefitial it is to remove the associated band for that feature. If the number is negative, it means that the band is benefitial. Its worth mentioning that this method has limitations, as the base algorithm a Light Gradient Boosting Machine is a very simple model in contrast with our architectures. Therefore, the effects of removing a given band on this model might differ dramatically for the other models.

[1] Ablation is the process of recurrently testing the same model on the same dataset, with the only difference being the removal of a given feature, for every feature in the dataset.

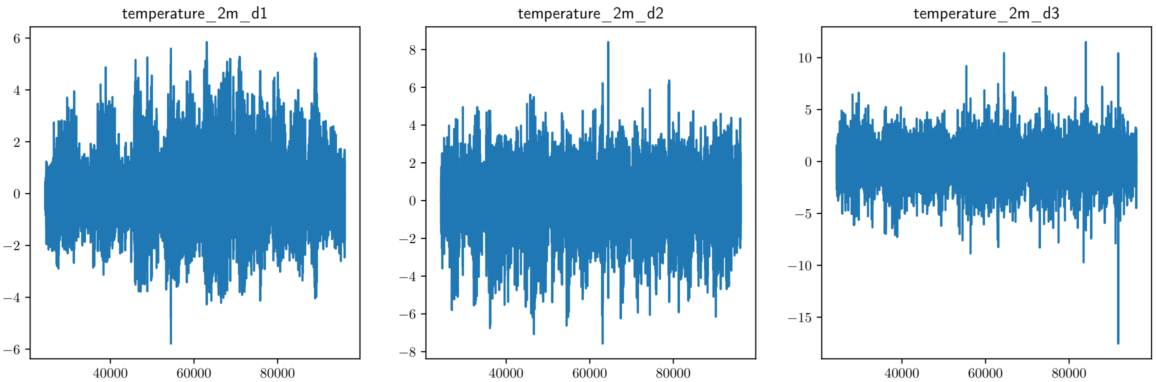

Figure 3: A plot of the temperature over time in our dataset after Wavelet decomposition

3 Results

For our performance metrics, we will assume RMSE and MAE. The difference between the two, for the purposes of this paper, is their tolerance to outlier data. MAE is more representative of the typical performance, while RMSE is outlier sensitive. So we could interpret a better MAE as more adapt for normal circumstances. Conversely, we can interpret RMSE as a good metric for how well the models can handle outlier data.

Table 4: Table with the results of the experiments, including the Proton model

run_title	**duration_hhmmss**	**mae**	**rmse**
Prophet baseline	03:44:00	35.705599	51.297253
Mean baseline	00:00:01	41.667505	55.593261
Zero baseline	00:00:00	87.795577	100.832967
MinMax + Mean	00:00:02	41.667505	55.593261
Exp1.a CNN-BiLSTM + Norm	00:01:27	39.738697	54.504961
Exp1.b CNN-xLSTM + Norm	01:40:47	36.616713	51.924452
Exp1.c CNN-BiLSTM-Tx + Norm	00:02:41	39.024715	52.396946
Exp1.d CNN-Tx + Norm	00:01:55	**35.202300**	**49.788010**
Exp1.a CNN-BiLSTM + Wavelet	00:02:03	38.347490	51.884933
Exp1.b CNN-xLSTM + Wavelet	02:44:48	37.643475	52.044102
Exp1.c CNN-BiLSTM-Tx + Wavelet	00:04:25	37.002768	**50.413288**
Exp1.d CNN-Tx + Wavelet	00:02:31	36.766987	51.588176
Exp1.a CNN-BiLSTM + Patch	00:00:55	42.927602	54.618952
Exp1.b CNN-xLSTM + Patch	00:11:38	43.010114	55.008874
Exp1.c CNN-BiLSTM-Tx + Patch	00:01:44	40.832303	54.211177
Exp1.d CNN-Tx + Patch	00:01:26	42.898806	54.757634
Exp2.a Decoder-Only Tx + Exp2 Standard	00:02:52	**31.805239**	**45.796124**
Exp2.b Decoder-Only Tx + Exp2 Wavelet	00:06:11	36.966030	**50.014923**
Exp2.c Encoder-Decoder Tx + Exp2 Standard	00:03:43	**30.480106**	**44.597015**
Exp2.d Kernel Tx + Kernel	00:03:32	**33.739584**	**48.833119**
Exp2.e iTransformer + Exp2 Standard	00:07:32	**30.210997**	**42.766888**

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